Birla Institute of Technology, Mesra
Off-Campus Jaipur
Minor Project – SEM-7

PROJECT SYNOPSIS

TripSolve: An Optimization-Grade Group Itinerary Planner for Jaipur Tourism

(Team Orienteering Problem with Time Windows, solved via OR-Tools, with LLM-assisted parsing and explainability)

1. Title of the Project

TripSolve: An Optimization-Grade Group Itinerary Planner for Jaipur Tourism (using TOPTW, OR-Tools, and LLM-Assisted Explainability)

2. Problem Statement

Planning a multi-stop tourist itinerary in a city like Jaipur is a genuinely hard combinatorial problem. A traveler, or a group of travelers, must decide which attractions to visit, in what order, while respecting real-world constraints:

- Each site's opening and closing hours (time windows)
- Realistic travel time between stops on actual roads, not straight-line distance
- For groups, the often-conflicting interests and priorities of different members

Where current tools fall short

- **LLM-only generation:** Most existing apps (Wanderlog, TripIt, and various LLM-based itinerary generators) prompt a language model to produce a "plausible-looking" day plan. Research such as the TravelPlanner benchmark (ICML 2024) has shown that pure LLM-based agents frequently violate hard constraints like time windows, because they pattern-match a reasonable-sounding sequence instead of solving the underlying optimization problem.
- **Single-user design:** When a group must share one route, most tools either ignore the conflict or let one person's preferences silently dominate, with no transparency into how the trade-off was made.

What this project addresses

- Generating itineraries grounded in real optimization over real-world constraints (opening hours, travel time, visit duration) rather than an LLM's guess
- Fairly balancing multiple travelers' preferences into a single shared route, with the reasoning behind the final route made transparent and explainable to the user

3. Methodology / Technology / Tools to be Used

The core problem is modeled as a Team Orienteering Problem with Time Windows (TOPTW): given a limited time budget and a set of attractions each with a location, visit duration, and opening/closing

window, the system selects and orders the subset of attractions that maximizes visitor value while respecting all time constraints. For a group, individual preference scores are combined into a shared multi-objective utility function so that no single member's interests dominate the route.

- **Backend & Optimization:** Python, Google OR-Tools (constraint programming solver for the TOPTW formulation), FastAPI (REST API layer)
- **Routing & Geospatial Data:** OSRM (Open Source Routing Machine) for real drive/walk times between Jaipur attractions, OpenStreetMap + Nominatim for geocoding, PostgreSQL with the PostGIS extension for geospatial storage and queries
- **AI / LLM Layer:** A self-hosted or Groq-hosted LLM (e.g., Llama 3.3) used strictly for two auxiliary tasks — parsing natural-language stop/preference input into structured data, and generating a plain-language explanation of why the solver produced a given route. The actual planning decision remains with the OR-Tools solver, not the LLM
- **Frontend:** Next.js with Leaflet.js for an interactive map showing draggable stops and the optimized route
- **Authentication:** JWT-based auth for saving and sharing group trips
- **Deployment:** Vercel (frontend) and Render / Fly.io (backend)

4. Expected Outcome

A working web application, seeded with real Jaipur attraction data (locations, opening hours, typical visit durations), that allows a single traveler or a group of travelers to input their available time and interests and receive a route that is:

- Provably time-window-feasible and near-optimal — rather than an LLM-generated guess
- A single shared route for groups, balancing multiple members' preferences
- Accompanied by a plain-language explanation of key trade-offs made (e.g., why one attraction was prioritized over another)

We expect to demonstrate measurable improvement over naive/manual planning in terms of total travel time and number of time-window violations avoided, along with a documented evaluation comparing the solver-based approach against a baseline pure-LLM itinerary generator on the same test scenarios. The project is also expected to result in a modular, city-agnostic backend architecture (Jaipur used as the seeded dataset for this synopsis) that could be extended to additional cities in future work.

5. Team

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